A Shortcut-Resistant and Domain-Adaptive Deep Learning Architecture for Peripheral Leukocyte Classification: Multi-Scale Preprocessing, Online Explainability Guardrails, and Agentic Clinical Verification

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October 2026

Abstract

Automated classification of peripheral white blood cells (leukocytes) speeds hematological screening and reduces manual review time in clinical laboratories. However, convolutional networks often lose accuracy when evaluated on out-of-distribution (OOD) blood smears collected with different cameras and stain preparations. These models also risk exploiting non-cellular background patterns (shortcut learning) while missing diagnostic cell morphology. To address both issues without large architectures or target-domain retraining, we introduce a shortcut-resistant and domain-adaptive classification framework based on a compact DenseNet121 backbone. The framework incorporates five components: (1) a five-stage Medical Enhanced Filter (MEF) that evens out cross-sensor exposure and blends dual-scale luminance contrast with bilateral smoothing and morphological edge operations; (2) a sequential channel-spatial attention module (*WBCAttention*) and parametric activation (*MedSwish*) that retain negative-domain chromatin gradients; (3) an online explainability callback (`XAIFocusMonitor`) paired with asymmetric foreground cropping and background noise to halt training if gradient attribution leaves the leukocyte body; (4) an inference-time adaptation pipeline that separates domain shift into Reinhard stain transfer, lightweight test-time augmentation (TTA), and target label-subspace projection; and (5) a clinical verification module combining foreground-masked Guided Grad-CAM (using 82nd-percentile dynamic alpha weighting) with a four-tier fault-tolerant multi-modal Large Language Model (LLM) agent. On the Raabin-WBC dataset, the 7.83M-parameter model reaches 98.53% in-distribution accuracy (0.9854 weighted F_1, 1.0000 F_1 on rare Basophils) with a median single-image latency of 18.5 ms on a consumer GPU (56.5 ms on CPU). On cross-device TestB smears, the adaptation pipeline raises accuracy to 89.05%, up from 56.96% for the unadapted five-class model and 73.90% after two-class subspace projection. That is a gain of 32.09 percentage points overall, of which 15.15 come from stain transfer and TTA alone.

Keywords: white blood cell classification, out-of-distribution generalization, test-time domain adaptation, shortcut learning prevention, explainable artificial intelligence (XAI), Guided Grad-CAM, multi-modal LLM verification, clinical MLOps.

1 Introduction

Microscopic examination of stained peripheral blood smears is standard practice for evaluating leukocyte morphology and differential counts. Quantifying the proportions and structure of the five primary white blood cell (WBC) subtypes (Neutrophil, Eosinophil, Basophil, Lymphocyte, and Monocyte) provides diagnostic evidence for bacterial infections, viral illnesses, hypersensitivity reactions, and hematological malignancies such as leukemia. Despite widespread laboratory automation, differential counting still requires manual inspection under a light microscope by a pathologist or medical technologist. This manual process takes roughly 15 to 20 minutes per slide, slowing workflow in high-volume laboratories and introducing inter-observer variation from human fatigue and subjective judgment.

Deep convolutional neural networks (CNNs) can automate this step with consistent, high-throughput results. They perform well on smears from the laboratory, microscope, and staining protocol used during training, but three problems stand in the way of clinical use. First, clinical smears vary across centers because of differences in optics, camera sensors (from laboratory CCDs to smartphone camera attachments), light sources, and stain chemistry. Models trained on a single source often drop sharply in accuracy when pre-

sented with out-of-distribution (OOD) images from another microscope. Second, models trained on unsegmented cell patches frequently learn background shortcuts: they rely on red blood cell density, slide illumination gradients, or stain residue, bypassing cytological markers such as nuclear lobation and cytoplasmic granulation. Third, natural class distributions are heavily skewed: common subtypes such as Neutrophils outnumber rare Basophils by more than twenty to one, which biases standard cross-entropy loss toward majority classes and yields unreliable minority predictions.

Most published leukocyte classification studies test only on single-domain data, so cross-device stability, interpretability, and latency go unreported. Stain normalization methods tend to rely either on uncalibrated optical density transforms that shift cytoplasmic colors or on large networks (over 100 million parameters) that are too slow for routine clinical systems. Explainable artificial intelligence (XAI) methods such as Grad-CAM usually appear as post-hoc figures in publications and play no part in training. A deployable system needs a lightweight model that attends to cell morphology during training, adjusts to new hardware at inference time without retraining, and checks its own heatmaps before a clinician sees the result.

This study presents a leukocyte classification pipeline built to resist shortcuts and adapt to new domains. Its contributions are:

1. **Five-Stage Medical Enhanced Filter (MEF):** A deterministic preprocessing sequence that uses percentile exposure clipping, Canny edge-weighted dual-scale CLAHE (4×4 nuclear and 8×8 cytoplasmic grids), bilateral filtering, elliptical morphological gradient blending, and selective Laplacian-of-Gaussian sharpening.

2. **Parameter-Efficient Architecture (*WBCAttention* and *MedSwish*):** A DenseNet121 backbone augmented with a channel-spatial attention gate (132,259 parameters) and a learnable activation function (*MedSwish*) that preserves negative-domain gradients on chromatin textures, with 7.83M total parameters, 98.53% accuracy on TestA, and a median single-image GPU latency of 18.5 ms, within 0.7 ms of the vanilla backbone.

3. **Two-Part Shortcut Learning Prevention:** Class-scaled foreground cropping and synthetic background noise paired with an online training callback (`XAIFocusMonitor`) that calculates Guided Grad-CAM foreground focus ratios (ρ) on cell masks, stopping training if attribution shifts toward background pixels.

4. **Retraining-Free Inference Adaptation:** A split of domain shift into covariate color shift and target label-space shift, showing that LAB Reinhard moment matching and 4-view test-time augmentation improve cross-hardware accuracy by +15.15 percentage points within the target subspace (and +32.09 percentage points overall, reaching 89.05% on TestB) without model updates.

5. **Foreground-Masked Guided Grad-CAM and Multi-LLM Verification:** An 82nd-percentile dynamic alpha heatmap visualization (α_{focus}) checked at inference time by a morphology-anchored, four-stage fault-tolerant LLM workflow (GPT-4o Vision → GPT-4o Text → Gemini 2.5 Flash → Rule-Based Fallback) served through a RESTful API.

Section 2 reviews peripheral leukocyte morphology, color spaces, and domain shift definitions. Section 3 discusses related work in leukocyte classification and stain adaptation. Section 4 describes the preprocessing filter, network architecture, and inference adaptation stack. Section 5 details the dataset, experimental benchmarks, ablation studies, and API performance. Section 6 concludes with limitations and directions for future work.

2 Fundamental Definitions, Concepts, and Preliminaries

The method draws on cell morphology, color spaces, domain shift, and attention-augmented CNNs. This section defines the terms used later.

2.1 Peripheral Leukocyte Morphology

Peripheral leukocytes are grouped into granulocytes (Neutrophils, Eosinophils, Basophils) and agranulocytes (Lymphocytes, Monocytes) based on nuclear shape and cytoplasmic granule staining under Romanowsky-Giemsa methods (Bain, 2005; Ceelie et al., 2007). *Neutrophils* (40–70% of circulating WBCs) have a segmented nucleus with 3 to 5 lobes connected by thin chromatin strands, surrounded by pale pink-violet cytoplasm. *Eosinophils* (1–4%) have a bilobed nucleus and coarse, orange-red granules. *Basophils* (< 1%) are the least common subtype, recognized by dense, dark purple-black granules that often obscure an S-shaped nucleus. *Lymphocytes* (20–40%) have a high nuclear-to-cytoplasmic ratio, a dense spherical nucleus, and a narrow rim of light blue cytoplasm. *Monocytes* (2–8%) are the largest subtype, with an indented or kidney-shaped nucleus and gray-blue cytoplasm that often contains small vacuoles (Bain, 2005).

2.2 Perceptual Color Spaces and Domain Shift Taxonomy

Standard RGB representations couple lightness with color, making pixel values sensitive to microscope lamp fluctuations and camera exposure differences (Reinhard et al., 2001). The CIE $L^*a^*b^*$ color space separates luminance $L^* \in [0, 100]$ from two opponent chromaticity axes, a^* (green–magenta) and b^* (blue–yellow). Modifying only L^* allows local contrast enhancement of nuclear chromatin without altering the purple and blue stain ratios encoded in (a^*, b^*) (Zuiderveld, 1994).

Let $\mathcal{D}_S = \{(x_i^S, y_i^S)\}_{i=1}^{N_S} \sim P_S(X, Y)$ denote a source training domain collected under laboratory setup S, and let $\mathcal{D}_T = \{(x_j^T, y_j^T)\}_{j=1}^{N_T} \sim P_T(X, Y)$ denote a target domain from unseen hardware T (Pan and Yang, 2010). Cross-laboratory transfer involves two main shifts:

1. **Covariate (Stain/Sensor) Shift:** $P_S(X) \neq P_T(X)$, while the underlying morphological relationship $P_S(Y \mid \phi(X)) = P_T(Y \mid \phi(X))$ remains valid under a domain-invariant feature mapping $\phi(\cdot)$.
2. **Target Label-Space Shift:** The target label set $\mathcal{Y}_T \subset \mathcal{Y}_S$ contains only a subset of the source classes $\mathcal{Y}_S = \{1, \ldots, C\}$. Standard softmax classifiers will assign probability mass to classes $\{c \notin \mathcal{Y}_T\}$ that do not exist in the target domain.

2.3 Dense Connectivity and Sequential Attention

In feed-forward CNNs, layer ℓ receives inputs only from layer $\ell-1$, which can slow gradient propagation in deep networks (Simonyan and Zisserman, 2015; He et al., 2016). Densely Connected Convolutional Networks (DenseNets) concatenate feature maps from all prior layers within each dense block (Huang et al., 2017):

$$\mathbf{X}_\ell = H_\ell\big([\mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_{\ell-1}]\big) \tag{1}$$

where $H_\ell(\cdot)$ applies batch normalization, non-linear activation, and 3×3 convolution. Reusing earlier features helps retain fine cytoplasmic edges alongside deeper nuclear representations while keeping the total parameter count relatively low.

To keep background pixels from dominating global pooling, the Convolutional Block Attention Module (CBAM) splits feature weighting into 1D channel attention $\mathbf{M}_c \in \mathbb{R}^{1\times1\times C}$ followed by 2D spatial attention $\mathbf{M}_s \in \mathbb{R}^{H\times W\times1}$ (Woo et al., 2018), so that informative channels and cell regions receive more weight.

2.4 Gradient Attribution and Shortcut Learning

Shortcut learning occurs when a network minimizes training loss by relying on non-biological image patterns, such as plasma background color, illumination falloff near the edges, or red blood cell packing, that correlate with class labels in the training set but do not hold across clinical centers (Geirhos et al., 2020; DeGrave et al., 2021). Gradient-weighted Class Activation Mapping (Grad-CAM) tracks spatial attribution by taking the gradient of class score y^c with respect to the final convolutional feature maps $\mathbf{A}^k \in \mathbb{R}^{u\times v}$ (Selvaraju et al., 2017). Guided Grad-CAM keeps only positive forward activations and positive backpropagated gradients (Springenberg et al., 2015), which gives a finer map that can resolve cellular substructures.

3 Literature Review and Related Work

Over the past twenty years, automated leukocyte analysis has moved from hand-crafted segmentation and texture descriptors to deep convolutional networks, and more recently to domain adaptation and explainable decision support.

3.1 Classical Image Processing and Machine Learning (2005–2015)

Early hematology pipelines relied on a sequence of cell segmentation, feature extraction, and shallow classifiers. Scotti (2005) analyzed leukocyte morphology for acute leukemia using low-pass background subtraction and contour descriptors, noting that fixed thresholds fail when slide illumination varies. Rezatofighi and Soltanian-Zadeh (2011) segmented cell nuclei using Gram-Schmidt orthogonalization with active contours and Support Vector Machines (SVMs), reporting 90.0% accuracy on 400 smear images at the expense of iterative contour computation. Su et al. (2014) combined 20 shape, color, and local-direction texture features to train a three-stage neural network; while accurate on clean slides, the hand-crafted thresholds broke down when cells touched adjacent red blood cells. Elen and Turan (2019) compared Decision Trees, k-Nearest Neighbors, SVMs, and Logistic Regression on geometric measurements, while Banik et al. (2020) combined K-means clustering in $L^*a^*b^*$ space with a small CNN. These classical methods depend on segmentation heuristics that do not generalize well across different laboratories.

3.2 Deep Convolutional and Transfer Learning Architectures (2016–2025)

Deep learning shifted automated hematology toward end-to-end representation learning (Esteva et al., 2019). Shahin et al. (2019) proposed WBCsNet and evaluated fine-tuned AlexNet and VGGNet backbones, reporting 96.1% accuracy on 2,551 images. Kutlu et al. (2020) and Shah et al. (2022) applied R-CNN and YOLO detectors to localize and classify leukocytes simultaneously, while Baydilli and Atila (2020) tested Capsule Networks (CapsNet) on small sample cohorts. Bozkurt (2021) and Girdhar et al. (2022) showed that DenseNet121 and custom CNNs reach 94–98% in-distribution accuracy on public benchmarks. Akalın and Yumuşak (2022) evaluated YOLOv5 and Detectron2 on the Raabin-WBC dataset (Kouzehkanan et al., 2022), obtaining 98.0% accuracy with a detection ensemble.

These studies report high in-distribution accuracy but share four limitations: (1) evaluations focus almost entirely on single-source test sets (such as Raabin-WBC TestA), overlooking the sharp drop seen on cross-device

splits (TestB); (2) large networks like VGG16 (138M parameters) require substantial memory and have high inference latency (Simonyan and Zisserman, 2015); (3) class imbalance is rarely handled in the loss function, leading to erratic performance on rare Basophils and Monocytes (Yeung et al., 2022); and (4) models operate as black boxes without built-in verification mechanisms.

3.3 Domain Shift, Stain Normalization, and Test-Time Adaptation

Accuracy loss across microscope setups is still a major obstacle to clinical adoption (Zhang et al., 2022; Esmaeilzadeh, 2024). Standard stain normalization aligns color distributions between a reference smear and target images. Reinhard et al. (2001) matched channel means and standard deviations in $L^*a^*b^*$ space. Macenko et al. (2009) factored optical density matrices via Singular Value Decomposition (SVD) to estimate separate hematoxylin and eosin vectors. In Giemsa-stained peripheral smears, however, uncalibrated SVD decomposition often flips stain channels when clear plasma dominates the image patch. At test time, Wang et al. (2019) and Putzu et al. (2025) showed that Test-Time Augmentation (TTA) reduces prediction variance on cross-domain cells, though heavy geometric or color transforms can distort cytoplasmic granules if applied without constraints.

3.4 Shortcut Learning, Explainability, and Clinical MLOps

Geirhos et al. (2020) and DeGrave et al. (2021) demonstrated that medical CNNs often achieve high validation scores by latching onto hospital-specific background marks or illumination patterns unrelated to pathology. Although Grad-CAM (Selvaraju et al., 2017) is commonly used to show model attention, most studies treat it only as a qualitative figure. Bassi et al. (2024) argued that explainability metrics should be incorporated into training to penalize spurious background correlations. From an engineering standpoint, Jackson et al. (2019), Bartels et al. (2022), and Abdullah et al. (2022) note that clinical adoption requires turning isolated analysis scripts into fault-tolerant, monitored services that report confidence and explanation quality.

Table 1 sets these studies against our approach. Only one prior study evaluates on cross-device data, and none handles imbalance with a focal loss, guards against shortcuts during training, or verifies its explanations.

4 Proposed Shortcut-Resistant and Domain-Adaptive System Architecture

The design goals are a small parameter budget, resistance to background shortcuts, and cross-hardware generalization without retraining. Figure 1 shows the training loop with its shortcut monitor and the inference pipeline with test-time adaptation and agentic verification.

4.1 Five-Stage Deterministic Medical Enhanced Filter (MEF)

To reduce cross-device illumination variation and optical noise before feature extraction, each resized input image $\mathbf{I} \in [0, 255]^{224\times224\times3}$ passes through a five-stage deterministic filter (Figure 2 and Algorithm 1).

Stage 1: Per-Channel Percentile Exposure Normalization. Microscope halogen and LED light sources produce different overall brightness levels. Each color channel $c \in \{R, G, B\}$ is linearly scaled between its 2nd and 98th empirical percentiles $(P_2^{(c)}, P_{98}^{(c)})$ to remove saturated specular highlights and darkened vignette corners:

$$I_c^{(1)}(u, v) = \mathrm{clip}\left(\frac{I_c(u, v) - P_2^{(c)}}{P_{98}^{(c)} - P_2^{(c)} + \epsilon} \cdot 255,\ 0,\ 255\right) \quad (2)$$

where $\epsilon = 10^{-6}$ prevents division by zero on uniform patches.

Stage 2: Edge-Weighted Dual-Scale CLAHE in CIE $L^*a^*b^*$ Space. The normalized RGB image $\mathbf{I}^{(1)}$ is converted to CIE $L^*a^*b^*$ coordinates (L, a, b). Applying contrast enhancement only to luminance L keeps the color ratios (a, b) intact. Because a single grid size cannot balance both fine nuclear chromatin and larger cytoplasmic regions, we run two Contrast Limited Adaptive Histogram Equalization (CLAHE) operators in parallel: a fine-scale operator $\mathcal{C}_{\mathrm{fine}}(L)$ with a 4×4 tile grid and clip limit $\tau_f = 3.0$, and a coarse-scale operator $\mathcal{C}_{\mathrm{coarse}}(L)$ with an 8×8 tile grid and clip limit $\tau_c = 2.0$. Let $E_L = \mathrm{Canny}(L; 30, 100) \in \{0, 1\}^{H\times W}$ be the binary edge map, and let $w_e = G_{\sigma=3}^{(11\times11)} * E_L \in [0, 1]^{H\times W}$ be its Gaussian-smoothed edge confidence map. The fused luminance channel L_{fused} is formed by convex blending:

$$\begin{aligned} L_{\mathrm{fused}} = {} & \mathcal{C}_{\mathrm{fine}}(L) \odot \big(0.5 + 0.5\, w_e\big) \\ & + \mathcal{C}_{\mathrm{coarse}}(L) \odot \big(0.5 - 0.5\, w_e\big) \end{aligned} \quad (3)$$

where $\odot$ denotes element-wise multiplication. Converting $(L_{\mathrm{fused}}, a, b)$ back to RGB gives $\mathbf{I}^{(2)}$.

Stage 3: Edge-Preserving Bilateral Filtering. Sensor noise in the cytoplasm is smoothed without blurring the cell boundaries using a bilateral filter $\mathbf{I}^{(3)} = \mathrm{Bilateral}(\mathbf{I}^{(2)}; d, \sigma_{\mathrm{color}}, \sigma_{\mathrm{space}})$:

$$\begin{aligned} \mathbf{I}^{(3)}(\mathbf{p}) = {} & \frac{1}{W_{\mathbf{p}}} \sum_{\mathbf{q}\in\Omega_{\mathbf{p}}} G_{\sigma_{\mathrm{space}}}\big(\|\mathbf{p} - \mathbf{q}\|_2\big) \\ & \times G_{\sigma_{\mathrm{color}}}\big(\|\mathbf{I}^{(2)}(\mathbf{p}) - \mathbf{I}^{(2)}(\mathbf{q})\|_2\big)\, \mathbf{I}^{(2)}(\mathbf{q}) \end{aligned} \quad (4)$$

with neighborhood diameter $d \in \{5, 9\}$ and color/spatial standard deviations $\sigma_{\mathrm{color}}, \sigma_{\mathrm{space}} \in [50, 65]$.

Table 1: Comparison of representative leukocyte classification studies and their evaluation settings.

Study (Author, Year)	Core Methodology	Dataset / Scale	Reported IND Acc.	Cross-Device OOD Eval.	Imbalance Focal Loss	Active Shortcut Guardrail	Agentic XAI & API
Rezatofighi and Soltanian-Zadeh (2011)	Gram-Schmidt + Snake + SVM	Custom (400 imgs)	90.00%	×	×	×	×
Shahin et al. (2019)	Custom WBCsNet / Transfer CNN	ALL-IDB / Custom (2,551)	96.10%	×	×	×	×
Banik et al. (2020)	$L^*a^*b^*$ K-Means + 5-Layer CNN	BCCD / LISC	98.63%	×	×	×	×
Baydilli and Atila (2020)	Capsule Networks (CapsNet)	LISC (263 imgs)	96.86%	×	×	×	×
Bozkurt (2021)	DenseNet121 Transfer Learning	BCCD (12,444 aug.)	94.10%	×	×	×	×
Akalin and Yumuşak (2022)	Hybrid YOLOv5 + Detectron2	Raabin-WBC	98.00%	×	×	×	×
Girdhar et al. (2022)	Custom 6-Layer CNN	Kaggle WBC (12,500)	98.55%	×	×	×	×
Putzu et al. (2025)	CNN + OOD-Filtered TTA	Cross-Dataset WBC	95.80%	✓	×	×	×
Proposed System (Ours)	**MEF + DenseNet121 + WBCAttention + MedSwish + Reinhard/TTA**	**Raabin-WBC (16,632 imgs: Train, TestA, TestB)**	**98.53%**	**✓(89.05% TestB; +32.09 pp)**	**✓**	**✓(`XAIFocusMonitor`)**	**✓(Guided Grad-CAM + Multi-LLM)**

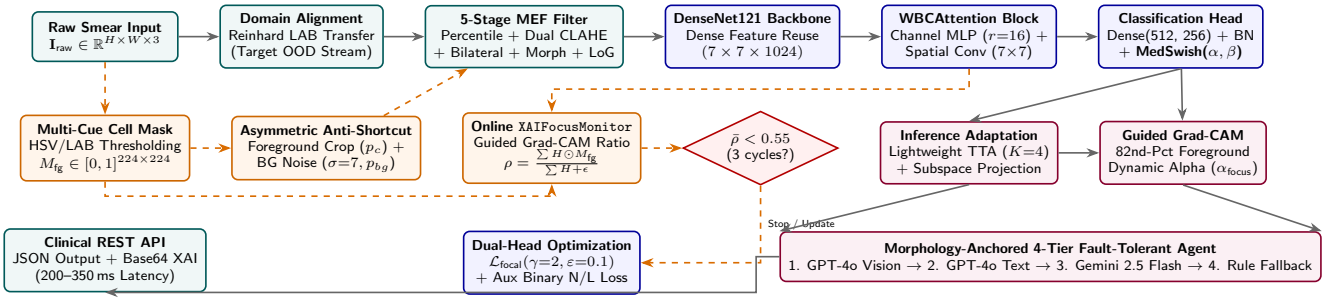


Figure 1: Block diagram of the proposed system: training loop with shortcut monitor (dashed orange) and inference pipeline with test-time adaptation and agentic verification (solid black).

Stage 4: Multi-Scale Elliptical Morphological Nucleus Emphasis. Nuclear lobation differentiates Neutrophils, Eosinophils, and Monocytes. We compute dual morphological gradients on the grayscale conversion $I_g = \text{Gray}(\mathbf{I}^{(3)})$ using elliptical structuring elements $k_3 \in \mathbb{R}^{3\times 3}$ and $k_7 \in \mathbb{R}^{7\times 7}$:

$$G_{\text{morph}} = 0.6\left(\delta_{k_3}(I_g) - \varepsilon_{k_3}(I_g)\right) + 0.4\left(\delta_{k_7}(I_g) - \varepsilon_{k_7}(I_g)\right) \tag{5}$$

where δ_k and ε_k denote grayscale dilation and erosion. The modulated image is computed as $\mathbf{I}^{(4)} = \text{clip}\big(\mathbf{I}^{(3)} \odot (1 + 0.2\, G_{\text{morph}}/255),\ 0,\ 255\big)$.

Stage 5: Selective Laplacian-of-Gaussian (LoG) Edge Sharpening. To sharpen chromatin edges without amplifying background noise, a combined edge mask $w_s \in [0,1]^{H\times W}$ is formed from the union of Canny edge detection and a thresholded Laplacian-of-Gaussian (LoG) filter, smoothed with a Gaussian kernel. High-boost convolution $\mathbf{I}_{\text{sharp}} = \mathbf{K}_{\text{sharp}} * \mathbf{I}^{(4)}$ (with center weight 9 and surround weights -1) is blended based on w_s with coefficient $\lambda_s \in [0.30, 0.35]$:

$$\mathbf{I}_{\text{MEF}} = \frac{1}{255}\ \text{clip}\Big(\mathbf{I}^{(4)} \odot \big(1 - \lambda_s w_s\big) + \mathbf{I}_{\text{sharp}} \odot \big(\lambda_s w_s\big),\ 0,\ 255\Big) \tag{6}$$

4.2 Hybrid Network Architecture: *WBCAttention* and *MedSwish*

The preprocessed tensor $\mathbf{I}_{\text{MEF}}$ enters an ImageNet-initialized DenseNet121 backbone ($7,037,504$ parameters), which produces a feature volume $\mathbf{F} \in \mathbb{R}^{7\times 7\times 1024}$.

Sequential Channel-Spatial *WBCAttention*. To prevent non-cellular background regions (erythrocytes and plasma) from distorting global pooling, $\mathbf{F}$ is refined through a sequential attention block. First, spatial dimensions are aggregated via parallel Global Average Pooling (GAP_s) and Global Max Pooling (GMP_s)

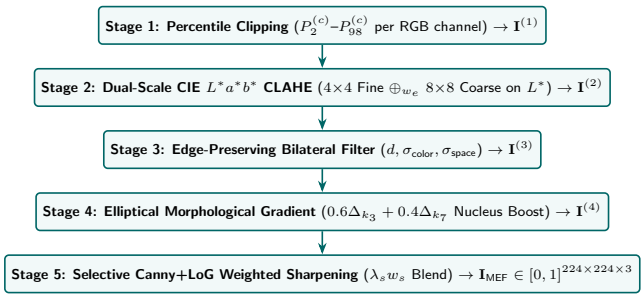


Figure 2: Processing stages of the deterministic Medical Enhanced Filter (MEF).

Algorithm 1: Deterministic 5-Stage Medical Enhanced Filter (MEF)

Input: Raw resized RGB image $\mathbf{I} \in [0,255]^{H\times W\times 3}$
Output: Normalized tensor $\mathbf{I}_{\text{MEF}} \in [0,1]^{H\times W\times 3}$

for channel $c \in \{R, G, B\}$ **do**
  Compute $P_2^{(c)}, P_{98}^{(c)} \leftarrow \text{Percentile}(I_c, \{2, 98\})$
  $I_c^{(1)} \leftarrow \text{clip}\big((I_c - P_2^{(c)})/(P_{98}^{(c)} - P_2^{(c)} + 10^{-6}) \cdot 255, 0, 255\big)$
end for
$(L, a, b) \leftarrow \text{RGB2LAB}(\mathbf{I}^{(1)})$
$L_f \leftarrow \text{CLAHE}_{4\times 4, 3.0}(L)$; $L_c \leftarrow \text{CLAHE}_{8\times 8, 2.0}(L)$
$w_e \leftarrow G_{\sigma=3}^{(11\times 11)} * (\text{Canny}(L, 30, 100)/255.0)$
$L_{\text{fused}} \leftarrow L_f \odot (0.5 + 0.5 w_e) + L_c \odot (0.5 - 0.5 w_e)$
$\mathbf{I}^{(2)} \leftarrow \text{LAB2RGB}(L_{\text{fused}}, a, b)$
$\mathbf{I}^{(3)} \leftarrow \text{BilateralFilter}(\mathbf{I}^{(2)}, d$=9, σ_c=65, σ_s=65)
$I_g \leftarrow \text{RGB2Gray}(\mathbf{I}^{(3)})$
$G_{\text{morph}} \leftarrow 0.6(\delta_{k_3}(I_g) - \varepsilon_{k_3}(I_g)) + 0.4(\delta_{k_7}(I_g) - \varepsilon_{k_7}(I_g))$
$\mathbf{I}^{(4)} \leftarrow \text{clip}\big(\mathbf{I}^{(3)} \odot (1 + 0.2 G_{\text{morph}}/255), 0, 255\big)$
$w_s \leftarrow G_{\sigma=2}^{(7\times 7)} * \text{BitwiseOr}(\text{Canny}(I_g), \text{LoG}(I_g))$
$\mathbf{I}_{\text{sharp}} \leftarrow \text{Conv2D}(\mathbf{I}^{(4)}, \mathbf{K}_{\text{sharp}})$
return $\text{clip}\big(\mathbf{I}^{(4)} \odot (1 - \lambda_s w_s) + \mathbf{I}_{\text{sharp}} \odot (\lambda_s w_s), 0, 255\big)/255.0$

into channel vectors $\mathbf{z}_{\text{avg}}, \mathbf{z}_{\max} \in \mathbb{R}^{1\times1\times C}$ ($C = 1024$). Both vectors traverse a shared two-layer bottleneck Multi-Layer Perceptron (MLP) with reduction ratio $r = 16$ ($\mathbf{W}_1 \in \mathbb{R}^{C\times(C/r)}$, ReLU activation, and $\mathbf{W}_2 \in \mathbb{R}^{(C/r)\times C}$, He-normal initialization):

$$\mathbf{M}_c(\mathbf{F}) = \sigma\Big(\mathbf{W}_2\ \mathrm{ReLU}(\mathbf{W}_1\mathbf{z}_{\text{avg}} + \mathbf{b}_1) + \mathbf{b}_2 + \mathbf{W}_2\ \mathrm{ReLU}(\mathbf{W}_1\mathbf{z}_{\max} + \mathbf{b}_1) + \mathbf{b}_2\Big) \quad (7)$$

$$\mathbf{F}' = \mathbf{F} \otimes \mathbf{M}_c(\mathbf{F}) \quad (8)$$

where $\sigma(\cdot)$ is the logistic sigmoid function and $\otimes$ denotes channel-wise broadcast multiplication. Second, channel-refined features $\mathbf{F}'$ are pooled across the channel axis via mean and max projections $\mathbf{S}_{\text{avg}}, \mathbf{S}_{\max} \in \mathbb{R}^{7\times7\times1}$, concatenated along the channel dimension, and convolved with a 7×7 spatial filter $\mathbf{f}^{7\times7}$:

$$\mathbf{M}_s(\mathbf{F}') = \sigma\Big(\mathbf{f}^{7\times7} * \big[\mathbf{S}_{\text{avg}};\ \mathbf{S}_{\max}\big]\Big) \quad (9)$$

$$\mathbf{F}_{\text{att}} = \mathbf{F}' \otimes \mathbf{M}_s(\mathbf{F}') \quad (10)$$

The module adds 132,259 parameters (1.69% of the model) and suppresses responses from surrounding erythrocytes.

Parametric *MedSwish* Activation. Following Global Average Pooling of $\mathbf{F}_{\text{att}}$ into a 1024-dimensional vector, classification proceeds through two fully connected layers (512 and 256 units) with Batch Normalization and Dropout ($p = 0.5$ and $p = 0.3$). ReLU ($\max(0, x)$) zeroes negative pre-activations, so subtle negative chromatin contrast features get no gradient. We use *MedSwish*, a parametric extension of the Swish family with learnable scalars α (initialized to 0.1) and β (initialized to 1.0):

$$\mathrm{MedSwish}(x;\alpha,\beta) = \underbrace{x\cdot\sigma(\beta x)}_{\substack{\text{parametric}\\ \text{Swish}}} \cdot \underbrace{\big(1+\alpha\tanh(x)\big)}_{\substack{\text{bounded odd}\\ \text{enhancement}}} \quad (11)$$

Because $\tanh(x) \in (-1, 1)$ is bounded and smooth, the multiplier $(1+\alpha\tanh(x))$ scales gradients in both positive and negative regimes while adding only 2 trainable parameters per dense layer (4 parameters total).

Dual-Head Output Topology. The 256-dimensional representation feeds two parallel heads: (1) the primary 5-class softmax head $\hat{\mathbf{p}} = \mathrm{Softmax}(\mathbf{W}_m\mathbf{h}+\mathbf{b}_m) \in \Delta^4$, and (2) an auxiliary binary sigmoid head $\hat{q} = \sigma(\mathbf{w}_{\text{aux}}^\top\mathbf{h} + b_{\text{aux}}) \in (0, 1)$ trained specifically on Neutrophil ($y_{\text{aux}} = 1$) versus Lymphocyte ($y_{\text{aux}} = 0$) samples to sharpen the decision boundary between the two classes evaluated on TestB. Table 2 details the layer dimensions and parameter counts.

4.3 Class-Balanced Optimization Objective

To address class imbalance without driving logit outputs into saturation, we combine a Label-Smoothed Class-Weighted Focal Loss ($\mathcal{L}_{\text{WBC-Focal}}$) on the primary head with sample-masked binary cross-entropy ($\mathcal{L}_{\text{aux}}$) on the auxiliary head. Given one-hot ground truth $\mathbf{y} \in \{0, 1\}^C$ and smoothing factor $\varepsilon = 0.1$, the smoothed target is $\tilde{y}_c = (1-\varepsilon)y_c + \varepsilon/C$. Defining $p_t = \sum_{c=1}^{C} y_c\hat{p}_c$ as the predicted probability for the ground-truth class, the primary loss is:

$$\mathcal{L}_{\text{WBC-Focal}}(\mathbf{y}, \hat{\mathbf{p}}) = -\alpha_t\big(1-p_t\big)^\gamma \times \sum_{c=1}^{C} w_c\,\tilde{y}_c\log\big(\mathrm{clip}(\hat{p}_c, \epsilon_p, 1-\epsilon_p)\big) \quad (12)$$

where $\gamma = 2.0$ down-weights easy majority examples, $\alpha_t = 0.25$ is the global scaling factor, $\epsilon_p = 10^{-7}$ ensures numerical stability, and $\mathbf{w} = [w_1, \ldots, w_C]^\top$ represents class-specific imbalance weights. The total training objective is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{WBC-Focal}} + \lambda_{\text{aux}}\mathbb{I}(y \in \{N, L\})\,\mathcal{L}_{\text{BCE}}(y_{\text{aux}}, \hat{q})$ with $\lambda_{\text{aux}} = 1.0$.

Table 2: Layer specifications and parameter counts of the proposed architecture.

Stage / Layer	Output Tensor Shape	Parameters	Role / Configuration
Input Tensor	(224, 224, 3)	0	MEF normalized float32
DenseNet121 Backbone	(7, 7, 1024)	7,037,504	Dense blocks 1-4 (ImageNet init)
`WBCAttentionBlock`	(7, 7, 1024)	132,259	Channel MLP (r=16) + 7×7 Spatial
GlobalAveragePooling2D	(1024)	0	Spatial aggregation
Dense 1 + BatchNorm 1	(512)	526,848	Feature projection + normalization
`MedSwish` 1 + Dropout 1	(512)	2	Learnable (α_1, β_1), Drop p=0.5
Dense 2 + BatchNorm 2	(256)	132,352	Bottleneck projection + normalization
`MedSwish` 2 + Dropout 2	(256)	2	Learnable (α_2, β_2), Drop p=0.3
Main Output (Softmax)	(5)	1,285	5-class leukocyte posterior $\hat{\mathbf{p}}$
Auxiliary Output (Sigmoid)	(1)	257	Neutrophil vs. Lymphocyte $\hat{q}$
Total Production Footprint	-	**7,830,509**	**7.83M params (≈94.3% < VGG16)**

4.4 Training-Time Shortcut Prevention and Online XAI Guardrail

To prevent the network from memorizing background erythrocytes or slide tint, we use a two-part training strategy:

Multi-Cue Unsupervised Cellular Mask Estimation (M_{fg}). Without requiring pixel-level ground-truth segmentation masks, a soft cellular foreground mask $M_{\text{fg}} \in [0, 1]^{H\times W}$ is estimated directly from HSV (H, S, V) and CIE $L^*a^*b^*$ (L, a, b) cues. Thresholds $\tau_S = \max(20, P_{60}(S))$ and $\tau_V = \min(245, P_{85}(V))$ isolate non-white plasma regions $B_{\text{nw}} = \mathbb{I}(S > \tau_S \vee V < \tau_V)$, and $\tau_a = P_{60}(a)$ and $\tau_b = P_{45}(b)$ isolate purple-stained nuclear chromatin $B_{\text{nuc}} = \mathbb{I}(a > \tau_a \wedge b < \tau_b)$. The union $B = B_{\text{nw}} \vee B_{\text{nuc}}$ undergoes morphological closing (5×5 kernel, 2 iterations), opening (1 iteration), contour filtering (retaining contours with area $> 0.01HW$), and Gaussian feathering ($\sigma_x = \sigma_y = 6.0$) normalized to $[0, 1]$.

Asymmetric Foreground Cropping and Background Randomization. During training, each sample undergoes stochastic perturbations using M_{fg}, scaled by class factors ($s_N = 0.3$ for Neutrophil, $s_L = 0.6$ for Lymphocyte, $s_c = 1.0$ otherwise): (1) *Foreground-Focused Cropping* ($p_{\text{crop}} = 0.20s_c$) crops around the bounding box of $\{M_{\text{fg}} > 0.2\}$ with a 22% margin and ±3% spatial jitter; (2) *LAB Stain Jitter* ($p_{\text{jitter}} = 0.30$) shifts luminance by $\Delta L \sim \mathcal{U}(-8, 8)$

and chromaticity by $\Delta a, \Delta b \sim \mathcal{U}(-6, 6)$; and (3) *Background Randomization* ($p_{\text{bg}} = 0.15 s_c$, mixing strength $\eta = 0.35$) replaces the exterior region $(1 - M_{\text{fg}})$ with a blurred ($\sigma \sim \mathcal{U}(2.5, 6.5)$), tinted, noise-injected ($\mathcal{N}(0, 7^2)$) synthetic background:

$$\begin{aligned}\mathbf{I}_{\text{aug}} = \mathbf{I} \odot \big(1 - \eta(1 - M_{\text{fg}})\big) \\ + \mathbf{I}_{\text{bg-synth}} \odot \big(\eta(1 - M_{\text{fg}})\big)\end{aligned} \tag{13}$$

Online `XAIFocusMonitor` Early-Stopping Callback. Every $K_e = 2$ epochs, Guided Grad-CAM heatmaps $H_i \in [0, 1]^{H \times W}$ are computed over a validation subset of $N_{\text{xai}} = 24$ images. The foreground focus ratio ρ_i measures the fraction of spatial attribution energy concentrated inside the cell mask $M_{\text{fg}}^{(i)}$:

$$\rho_i = \frac{\sum_{u=1}^{H} \sum_{v=1}^{W} H_i(u, v)\, M_{\text{fg}}^{(i)}(u, v)}{\sum_{u=1}^{H} \sum_{v=1}^{W} H_i(u, v) + \epsilon}, \tag{14}$$

$$\bar{\rho} = \frac{1}{N_{\text{xai}}} \sum_{i=1}^{N_{\text{xai}}} \rho_i$$

If the mean focus ratio falls below $\bar{\rho} < \tau_{\text{xai}} = 0.55$ for $P_{\text{xai}} = 3$ consecutive evaluation cycles, `XAIFocusMonitor` halts training early to prevent the model from learning background shortcuts.

4.5 Retraining-Free Inference-Time Domain Adaptation

When evaluating on target domain $\mathcal{D}_T$ without retraining, domain shift is addressed in three steps:

1. **Stain Alignment via LAB Moment Matching (Reinhard):** Reference means $\boldsymbol{\mu}_{\text{ref}} = [148.60, 169.30, 105.97]^\top$ and standard deviations $\boldsymbol{\sigma}_{\text{ref}} = [41.13, 9.01, 6.67]^\top$ are calculated in CIE $L^*a^*b^*$ space from $N_{\text{ref}} = 1,200$ source smears. For any target image $\mathbf{I}_T$ with statistics $(\boldsymbol{\mu}_T, \boldsymbol{\sigma}_T)$, each channel $k \in \{L^*, a^*, b^*\}$ is aligned before MEF filtering:

$$\tilde{I}_{T,k} = \text{clip}\left(\frac{I_{T,k} - \mu_{T,k}}{\sigma_{T,k} + \epsilon}\, \sigma_{\text{ref},k} + \mu_{\text{ref},k},\ 0,\ 255 \right) \tag{15}$$

2. **Test-Time Augmentation (TTA):** From the normalized tensor $\mathbf{X}_0 = \text{MEF}(\tilde{\mathbf{I}}_T)$, we form $M = 4$ views $\{\mathcal{T}_m(\mathbf{X}_0)\}_{m=1}^{4}$: identity, horizontal reflection, brightness scaling ($\times 1.05$), and $\pm 5^\circ$ planar rotation. The ensemble prediction averages their outputs:

$$\bar{\mathbf{p}}(\mathbf{X}_0) = \frac{1}{M} \sum_{m=1}^{M} f_{\boldsymbol{\theta}}\big(\mathcal{T}_m(\mathbf{X}_0)\big) \tag{16}$$

3. **Target Label-Subspace Projection:** When evaluating on a target benchmark whose label support is restricted to a subset $\mathcal{Y}_T \subset \{1, \ldots, C\}$ (such as Raabin-WBC TestB, where only Neutrophil and Lymphocyte annotations exist), the 5-class posterior $\bar{\mathbf{p}}$ is projected onto the active probability simplex $\Delta^{|\mathcal{Y}_T|-1}$ via Bayes' rule:

$$\hat{p}_{\text{proj}}(y = c \mid \mathbf{X}_0) = \frac{\bar{p}_c(\mathbf{X}_0)\, \mathbb{I}(c \in \mathcal{Y}_T)}{\sum_{j \in \mathcal{Y}_T} \bar{p}_j(\mathbf{X}_0) + \epsilon} \tag{17}$$

Section 5 reports both raw 5-class metrics and subspace-projected 2-class metrics to separate the contribution of visual stain and TTA adaptation (+15.15 pp) from label-subspace alignment (+16.94 pp).

4.6 Foreground-Masked Guided Grad-CAM and Fault-Tolerant Agentic Verification

To provide interpretable predictions and audit spatial attention at inference time, we combine foreground-masked Guided Grad-CAM with a four-tier LLM verification chain (Algorithm 2).

Guided Grad-CAM with 82nd-Percentile Foreground Dynamic Alpha (α_{focus}). Standard Grad-CAM heatmaps have coarse spatial resolution (7×7) and can bleed activation into surrounding plasma. Given predicted class $c^* = \arg\max_c \hat{p}_c$ and final convolutional feature maps $\mathbf{A}^k \in \mathbb{R}^{7 \times 7}$, Guided Grad-CAM gradients retain only positive forward activations and positive backward gradients:

$$\mathbf{g}_k^{\text{guided}} = \mathbb{I}\big(\mathbf{A}^k > 0\big) \odot \mathbb{I}\left(\frac{\partial y^{c^*}}{\partial \mathbf{A}^k} > 0\right) \odot \frac{\partial y^{c^*}}{\partial \mathbf{A}^k} \tag{18}$$

$$w_k = \frac{1}{49} \sum_{i=1}^{7} \sum_{j=1}^{7} g_{k,i,j}^{\text{guided}}$$

The raw map $H_{\text{raw}} = \max\big(0, \sum_k w_k \mathbf{A}^k\big)$ is normalized to $[0, 1]$, bilinearly upsampled to 224×224, smoothed with a Gaussian kernel ($\sigma = 2.0$), and damped outside the cell mask M_{fg}:

$$\begin{aligned} H_{\text{fg}}(u, v) &= H_{\text{smooth}}(u, v) \cdot \big(0.05 + 0.95\, M_{\text{fg}}(u, v)\big), \\ \tilde{H}(u, v) &= \frac{H_{\text{fg}}(u, v)}{\max_{u', v'} H_{\text{fg}}(u', v') + \epsilon} \end{aligned} \tag{19}$$

To isolate the most prominent cellular features, we apply power-law gamma sharpening $H_{\text{focus}} = \tilde{H}^{1.6}$ and set an adaptive threshold τ_f at the 82nd percentile of active foreground pixels $\Omega_{\text{fg}} = \{(u, v) \mid M_{\text{fg}}(u, v) > 0.15\}$:

$$\tau_f = \text{clip}\Big(P_{82}\big(H_{\text{focus}}\,|_{\Omega_{\text{fg}}}\big),\ 0.45,\ 0.90\Big) \tag{20}$$

$$\begin{aligned} \alpha_{\text{focus}}(u, v) = 0.78 \cdot M_{\text{fg}}(u, v) \\ \times \text{clip}\left(\frac{H_{\text{focus}}(u, v) - \tau_f}{1.0 - \tau_f + \epsilon},\ 0.0,\ 1.0\right) \end{aligned} \tag{21}$$

The visualization overlays the Jet-colormapped attribution $\mathbf{C}_{\text{jet}}(\tilde{H})$ onto the dimmed cellular base $\mathbf{I}_{\text{base}} = \mathbf{I}_{\text{orig}} \odot (0.55 + 0.45\, M_{\text{fg}})$ using α_{focus}.

Morphology-Anchored Fault-Tolerant Multi-LLM Verification. The overlay image $\mathbf{I}_{\text{XAI}}$, predicted class c^*, confidence $\hat{p}_{c^*}$, and class-specific cytological anchor $\mathcal{M}_{c^*}$ (encoding expected nuclear lobation, chromatin density, and cytoplasmic granule traits from Table 3) are passed to a four-tier fallback chain:

1. **Tier 1 (Primary Multimodal VLM):** OpenAI `GPT-4o` inspects $\mathbf{I}_{\text{XAI}}$ alongside $\mathcal{M}_{c^*}$ and checks whether high-attribution zones fall on the expected nuclear or cytoplasmic structures or leak into background artifacts.
2. **Tier 2 (Text-Only Fallback):** If image input is rejected or times out, `GPT-4o` generates a text review from $(\hat{\mathbf{p}}, \mathcal{M}_{c^*})$ and flags the image for manual inspection.
3. **Tier 3 (Alternative Multimodal VLM):** If the primary provider is unreachable, the system fails over to Google `Gemini 2.5 Flash` ($T = 0.2$) with identical multimodal grounding.
4. **Tier 4 (Rule-Based Fallback):** In air-gapped or offline setups, a deterministic lookup returns the morphological reference description and confidence tier.

Table 3: Cytological morphology anchors ($\mathcal{M}_c$) supplied to the multi-modal LLM verifier.

Subtype (c)	Expected Cytological Focus Anchor ($\mathcal{M}_c$) vs. Prohibited Shortcut
Neutrophil	Multi-lobed nucleus (3–5 lobes connected by thin filaments) and pale pink-violet fine granules; must ignore surrounding erythrocytes.
Lymphocyte	Large, round, hyperchromatic nucleus occupying most of cell volume with narrow basophilic cytoplasmic rim; focus on N:C ratio.
Monocyte	Indented kidney/horseshoe-shaped nucleus and abundant gray-blue vacuolated cytoplasm; focus on nuclear indentation and cytoplasmic texture.
Eosinophil	Bilobed nucleus and coarse, bright orange-red acidophilic cytoplasmic granules; must not rely on background plasma staining.
Basophil	Coarse, deeply stained purple-black basophilic granules overlapping an irregular nuclear outline; focus on dense granule clusters.

Algorithm 2: Foreground-Masked Guided Grad-CAM & Fault-Tolerant Agentic Audit

Input: Target smear $\mathbf{I}$, model $f_{\boldsymbol{\theta}}$, anchors $\{\mathcal{M}_c\}_{c=1}^{5}$
Output: JSON payload $\{c^*, \hat{p}_{c^*}, \mathbf{I}_{\text{XAI}}, \text{report } \mathcal{R}, \text{provider}\}$

$\mathbf{X} \leftarrow \text{MEF}(\mathbf{I})$; $\hat{\mathbf{p}} \leftarrow f_{\boldsymbol{\theta}}(\mathbf{X})$; $c^* \leftarrow \arg\max_c \hat{p}_c$
Compute Guided Grad-CAM H_{raw} at deepest Conv2D layer via Eq. (18)
Estimate cellular mask $M_{\text{fg}} \leftarrow \text{EstimateForegroundMask}(\mathbf{I})$
Compute foreground-damped $\tilde{H}$ (Eq. 19) and 82nd-pct α_{focus} (Eqs. 20–21)
$\mathbf{I}_{\text{XAI}} \leftarrow (1 - \alpha_{\text{focus}}) \odot \mathbf{I}_{\text{base}} + \alpha_{\text{focus}} \odot \mathbf{C}_{\text{jet}}(\tilde{H})$
Construct prompt $\mathcal{P} \leftarrow \text{BuildPrompt}(c^*, \hat{p}_{c^*}, \mathcal{M}_{c^*})$
try $\mathcal{R} \leftarrow \text{GPT4o_Vision}(\mathcal{P}, \mathbf{I}_{\text{XAI}})$, provider $\leftarrow$ "OpenAI GPT-4o"
catch VisionError: **try** $\mathcal{R} \leftarrow \text{GPT4o_Text}(\mathcal{P})$, provider $\leftarrow$ "GPT-4o (Text)"
catch PrimaryAPIError: **try** $\mathcal{R} \leftarrow \text{Gemini25Flash}(\mathcal{P}, \mathbf{I}_{\text{XAI}})$, provider $\leftarrow$ "Gemini 2.5 Flash"
catch AllProvidersFailed: $\mathcal{R} \leftarrow \text{RuleFallback}(c^*, \hat{p}_{c^*})$, provider $\leftarrow$ "Rule-based"
return $\{c^*, \hat{p}_{c^*}, \hat{\mathbf{p}}, \text{Base64}(\mathbf{I}_{\text{XAI}}), \mathcal{R}, \text{provider}\}$

5 Experimental Findings and Evaluation

We evaluate the framework in five scenarios: backbone comparison and latency, subtype classification on in-distribution data, cross-hardware adaptation on out-of-distribution data, preprocessing ablations, and REST API deployment.

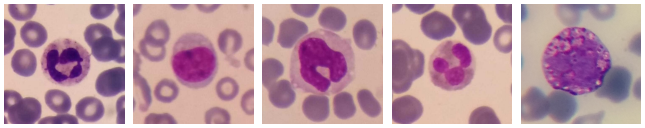
5.1 Dataset Description and Partitioning

All experiments use the open-access Raabin-WBC benchmark released by Tehran University (Kouzehkanan et al., 2022). The dataset contains 16,632 Giemsa-stained peripheral blood smear images divided into three disjoint splits (Table 4 and Figure 3):

- **Train Split ($n = 10,174$):** Stratified into 8,650 training images (85%) and 1,524 validation images (15%) across all five leukocyte subtypes, collected with an Olympus CX18/CX21 microscope paired with a Samsung Galaxy S5 adapter and a Canon EOS digital camera.
- **TestA Split ($n = 4,339$, In-Distribution / IND):** Held-out test split collected under the same optical and staining distribution as the training data. The split has substantial class imbalance, ranging from 2,660 Neutrophils (61.3%) to 89 Basophils (2.05%).
- **TestB Split ($n = 2,119$, Out-of-Distribution / OOD):** Cross-device generalization split collected with a different microscope lens and camera sensor. TestB contains labels only for *Neutrophil* (1,971 images) and *Lymphocyte* (148 images) subtypes.

Table 4: Sample distribution across training, validation, in-distribution (TestA), and out-of-distribution (TestB) splits of the Raabin-WBC dataset.

Leukocyte Subtype	Train (85%)	Val (15%)	TestA (IND)	TestB (OOD)	Total
Basophil	181	31	89	0	301
Eosinophil	632	111	322	0	1,065
Lymphocyte	2,063	364	1,034	148	3,609
Monocyte	477	84	234	0	795
Neutrophil	5,297	934	2,660	1,971	10,862
Total Samples	**8,650**	**1,524**	**4,339**	**2,119**	**16,632**



(a) Neutrophil (b) Lymphocyte (c) Monocyte (d) Eosinophil (e) Basophil

Figure 3: Representative Giemsa-stained peripheral leukocyte subtypes from the Raabin-WBC dataset showing differences in nuclear shape and cytoplasmic granules.

5.2 Implementation Details and Experimental Setup

All training and benchmarking ran on a workstation with an NVIDIA GeForce RTX 4070 Ti SUPER GPU (16 GB GDDR6X VRAM), an AMD Ryzen 7 7800X3D 8-core processor, and 32 GB DDR5 RAM running Windows 11 with WSL2 CUDA acceleration, Python 3.10+, TensorFlow/Keras, and OpenCV 4.8. Every backbone was trained on the same two-phase schedule with a fixed random seed ($s = 42$) and batch size $B = 16$:

- **Phase 1 (Warm-Up Feature Extraction, 15 Epochs):** Pre-trained ImageNet backbone weights are frozen; only the `WBCAttentionBlock`, `MedSwish` scalars (α, β), Batch Normalization, and dual classification heads are optimized using Adam (lr $= 1 \times 10^{-3}$).
- **Phase 2 (Full-Network Fine-Tuning, 15 Epochs):** All backbone layers are unfrozen and optimized end-to-end with Adam at a conservative learning rate (lr $= 1 \times 10^{-5}$) governed by `ReduceLROnPlateau` and online `XAIFocusMonitor` ($\tau_{\text{xai}} = 0.55$, patience 3).

5.3 Evaluation Metrics Formulations

Given true positives (TP_c), false positives (FP_c), and false negatives (FN_c) for class $c \in \{1, \ldots, C\}$ with support N_c ($N = \sum_{c=1}^{C} N_c$), we evaluate per-class Precision (Prec_c), Recall (Rec_c), $F_{1,c}$, unweighted Macro-F_1 (which weights rare Basophils equally with Neutrophils), and support-weighted F_1:

$$\text{Prec}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad \text{Rec}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \tag{22}$$

$$F_{1,c} = \frac{2\,\text{Prec}_c\,\text{Rec}_c}{\text{Prec}_c + \text{Rec}_c}, \quad \text{Macro-}F_1 = \frac{1}{C}\sum_{c=1}^{C} F_{1,c} \tag{23}$$

$$\text{Weighted-}F_1 = \sum_{c=1}^{C} \frac{N_c}{N} F_{1,c}$$

5.4 Scenario 1: Cross-Backbone Representation and Latency Benchmark

We first evaluated five transfer learning backbones (VGG16 (Simonyan and Zisserman, 2015), ResNet50V2 (He et al., 2016), MobileNetV2 (Sandler et al., 2018), EfficientNetB0 (Tan and Le, 2019), and Vanilla DenseNet121 (Huang et al., 2017)) against the proposed model under the same MEF preprocessing and two-phase training protocol. Table 5 reports validation metrics, parameter counts, inference latencies, and per-class validation F_1 scores. Figure 4 shows the training and validation convergence curves.

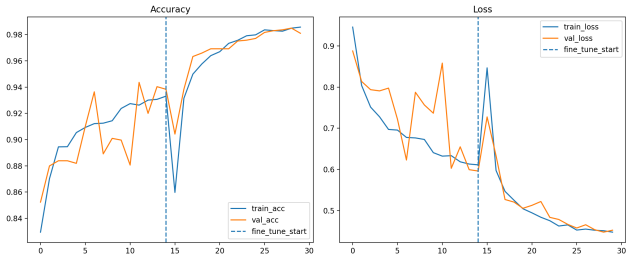


Figure 4: Two-phase curriculum training and validation accuracy/loss trajectories of the proposed architecture. The vertical dashed line at epoch 15 marks the transition from frozen-backbone warm-up (Phase 1) to end-to-end fine-tuning (Phase 2).

The mobile architectures (MobileNetV2 and EfficientNetB0) do worst on Monocytes, the most morphologically subtle class (F_1 of 0.8727 and 0.8589), which suggests that depthwise-separable convolutions have limited capacity for fine chromatin texture. Vanilla DenseNet121 beats ResNet50V2 by 0.99% Macro-F_1 with less than a third of the parameters (7.70M vs. 24.75M); we attribute this to dense feature concatenation. The proposed model's row reports TestA, not validation, scores (98.53% accuracy, 0.9755 Macro-F_1), so it cannot be ranked against the validation rows. On TestA, where Table 8 has every backbone, vanilla DenseNet121 and the proposed model are within 0.04 pp (98.57% vs. 98.53%). All six models were timed under the same protocol (table note). The attention block and MedSwish cost little: the proposed model's median latency is 18.5 ms on the GPU and 56.5 ms on the CPU, against 17.8 ms and 56.8 ms for vanilla DenseNet121. Its GPU tail is wider, with a 95th-percentile latency of 61.4 ms against 48.1 ms. Both DenseNet121 variants are the slowest of the six models on both devices. Execution mode matters more than the added blocks: without graph compilation, the GPU medians rise to 229.8 ms for vanilla DenseNet121 and 241.5 ms for the proposed model.

5.5 Scenario 2: In-Distribution Subtype Granularity on TestA

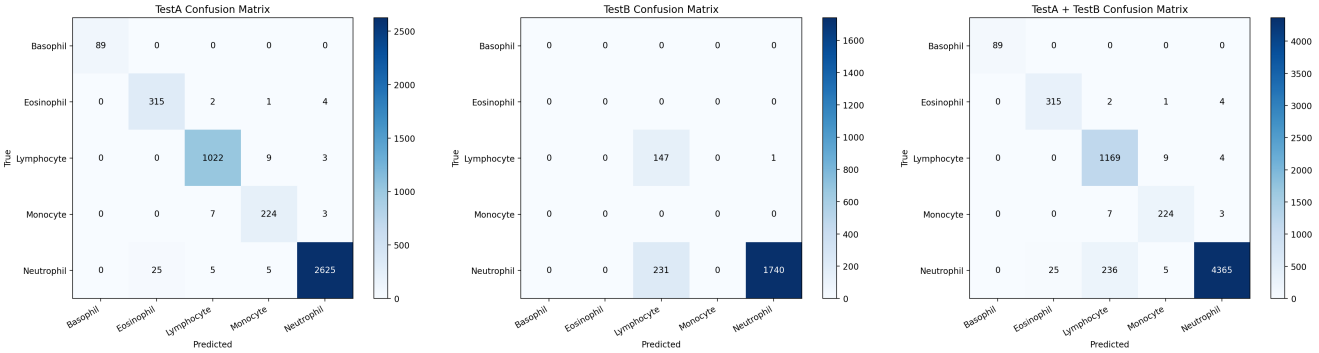
Table 6 reports per-class Precision, Recall, and F_1-Score on TestA ($n = 4,339$). Confusion matrices for TestA, TestB, and the combined evaluation set ($n = 6,458$) appear in Figure 5.

Basophils make up only 2.05% of TestA (89 samples), yet the model classifies all of them correctly (precision and recall of 1.0000), consistent with class-weighted focal loss and label smoothing keeping the

Table 5: Scenario 1: Transfer learning backbones on the validation split ($n = 1,524$), with parameter counts, single-image inference latency, and per-class F_1 scores. Bold marks the best baseline in each column. The proposed model's accuracy and F_1 scores are on TestA (see note); its latencies use the same protocol as the baselines.

Architecture	Params (M)	Val. Acc. (%)	Macro-F_1	Median Latency (ms)		Per-Class Validation F_1-Score				
				GPU	CPU	Basophil	Eosinophil	Lymphocyte	Monocyte	Neutrophil
VGG16 (Simonyan and Zisserman, 2015)	15.11	98.56	0.9724	**2.8**	34.4	1.0000	**0.9799**	0.9821	0.9067	0.9935
ResNet50V2 (He et al., 2016)	24.75	98.17	0.9704	6.0	30.7	1.0000	0.9600	0.9803	0.9222	0.9897
MobileNetV2 (Sandler et al., 2018)	3.05	97.90	0.9577	5.5	**21.0**	1.0000	0.9435	0.9803	0.8727	0.9919
EfficientNetB0 (Tan and Le, 2019)	4.84	97.05	0.9418	8.1	32.0	0.9730	0.9121	0.9818	0.8589	0.9833
DenseNet121 (Vanilla) (Huang et al., 2017)	7.70	**98.89**	**0.9803**	17.8	56.8	1.0000	0.9672	**0.9916**	**0.9494**	**0.9936**
Proposed (DenseNet121 + WBCAttention + MedSwish)	7.83	98.53*	0.9755*	18.5	56.5	1.0000*	0.9517*	0.9874*	0.9471*	0.9915*

*On held-out TestA ($n = 4,339$; Table 6), not the validation split.
Latency: batch size 1, float32, compiled graph execution (TensorFlow 2.21, Keras 3.14), 10 warm-up passes; median of 200 runs on an RTX 4070 Ti SUPER (GPU) and 100 runs on a Ryzen 7 7800X3D (CPU).



(a) TestA (In-Distribution, n=4,339) **(b)** TestB (OOD, n=2,119) **(c)** Combined Evaluation (n=6,458)

Figure 5: Confusion matrices of the proposed framework across (a) TestA in-distribution, (b) TestB cross-device out-of-distribution, and (c) combined evaluation splits.

Table 6: Scenario 2: Per-class Precision, Recall, and F_1-Score on the in-distribution TestA split ($n = 4,339$) and Combined split ($n = 6,458$).

Subtype	TestA In-Distribution ($n = 4,339$)				Combined ($n = 6,458$)	
	Precision	Recall	F_1-Score	Support	F_1-Score	Support
Basophil	**1.0000**	**1.0000**	**1.0000**	89	**1.0000**	89
Eosinophil	0.9265	0.9783	0.9517	322	0.9517	322
Lymphocyte	0.9865	0.9884	0.9874	1,034	0.9006	1,182
Monocyte	0.9372	0.9573	0.9471	234	0.9471	234
Neutrophil	0.9962	0.9868	0.9915	2,660	0.9692	4,631
Macro Avg	0.9693	0.9822	0.9755	4,339	0.9537	6,458
Weighted Avg	**0.9856**	**0.9853**	**0.9854**	**4,339**	**0.9554**	**6,458**

majority classes from swamping rare ones. Neutrophil and Lymphocyte reach F_1 scores of 0.9915 and 0.9874. Eosinophil precision is lower (0.9265, $F_1 = 0.9517$) because 19 heavily granulated Neutrophils were classified as Eosinophils (Figure 5a), a known morphological ambiguity in granular smears.

5.6 Scenario 3: Out-of-Distribution (OOD) Generalization and Adaptation Ablation

Table 7 adds the adaptation steps one at a time on TestB ($n = 2,119$), so the gain from label-subspace alignment can be read separately from the gain from color transfer. Table 8 gives the same TestA, TestB, and Combined accuracies for the other backbones.

The unadapted 5-class model scores 56.96% on TestB because sensor color differences lead it to assign Neutrophils to granulocyte classes that TestB does not contain (Eosinophil and Monocyte). Projecting logits onto the two-class subspace ($\{N, L\}$) gives an unadapted binary baseline of 73.90% (+16.94 pp). From there, CIE $L^*a^*b^*$ Reinhard stain transfer adds 12.56 pp (to 86.46%), the largest visual gain, which suggests that chromaticity drift between laboratories is the main component of covariate shift. 4-view TTA adds another 2.59 pp, for 89.05% TestB accuracy (0.9994 Neutrophil precision, 0.9932 Lymphocyte recall; Figure 5b), and raises TestA accuracy from 97.46% to 98.53%. Among the other backbones (Table 8), MobileNetV2 drops to 50.07% on TestB, and Vanilla DenseNet121 falls from 98.57% on TestA to 73.24%.

5.7 Scenario 4: Preprocessing Pipeline Ablation

We ran the trained network with four preprocessing variants (`eval_prep_v1` through `eval_prep_v4`) on TestA, TestB, and the Combined split (Table 9).

The ablation shows three effects:

1. **Spatial Scale Trade-Off (`v1` vs. `v2`):** Expanding the CLAHE grid to a single 8×8 scale (`v2`) smooths sensor noise on TestB (+2.27 pp) but reduces chromatin contrast on TestA (−0.42 pp). The Canny edge-weighted dual-scale blend (Equation 3) balances both scales.
2. **Loss of Cytoplasmic Texture (`v3`):** Adding white top-hat and black bottom-hat filters to `v2` (`v3`) reduces accuracy by 2.81 pp on TestA and 10.34 pp on TestB because the 11×11 morphological filter removes subtle cytoplasmic texture needed to separate

Table 7: Scenario 3A: Incremental ablation of retraining-free inference-time adaptation across TestA (IND), TestB (OOD), and Combined splits, separating Label-Space Shift from Covariate Stain Shift.

Stage	Inference Configuration	Shift Addressed	TestA Acc. (%)	TestB Acc. (%)	TestB Δ (pp)	Combined Acc. (%)
0a	Unprojected 5-Class Baseline (`densenet_fair`, plain CCE)	None (5-class raw)	97.44	54.55	-	83.40
0b	Unprojected 5-Class Shortcuts-Resistant Baseline (MEF v1 only)	None (5-class raw)	97.46	56.96	Baseline	84.17
1	+ Target Subspace Prior Projection ($\mathcal{Y}_T = \{N, L\}$, Eq. 17)	Label-Space Shift	97.46	73.90	+16.94	89.73
2	+ CIE $L^*a^*b^*$ Reinhard Stain Transfer (Eq. 15)	Covariate Stain Shift	97.99	86.46	+12.56 (Σ+29.50)	94.21
3	+ Lightweight 4-View TTA Ensemble (Eq. 16, Final Proposed)	Prediction Variance	**98.53**	**89.05**	+2.59 (Σ+32.09)	**95.42**

Methodological Note: Within the valid 2-class target subspace of TestB (Stage 1 → Stage 3), pure visual adaptation (Reinhard + TTA) yields a net **+15.15 pp** OOD gain (73.90% → 89.05%).

Table 8: Scenario 3B: Cross-backbone OOD generalization comparison across TestA, TestB, and Combined splits.

Model Architecture	Params (M)	TestA (%)	TestB OOD (%)	Combined (%)
VGG16 (Simonyan and Zisserman, 2015)	138.36*	98.39	**89.71**	**95.54**
ResNet50V2 (He et al., 2016)	24.75	97.23	74.47	89.76
MobileNetV2 (Sandler et al., 2018)	3.05	95.92	50.07	80.88
EfficientNetB0 (Tan and Le, 2019)	4.84	96.27	61.54	84.87
DenseNet121 (Vanilla) (Huang et al., 2017)	7.70	**98.57**	73.24	90.26
Proposed System (Final)	**7.83**	98.53	89.05	95.42

*Full VGG16 architecture footprint (15.11M convolutional backbone + dense top).

Table 9: Scenario 4: Ablation of preprocessing pipeline variants on the trained model across TestA (IND), TestB (OOD, with subspace projection), and Combined splits.

ID	Preprocessing Configuration	TestA (%)	TestB (%)	Comb. (%)
v1	**MEF Original (Dual-Scale 4×4/8×8 CLAHE + Morph + LoG)**	**98.41**	85.65	94.22
v2	Single-Scale Adaptive CLAHE (8×8 grid, σ_L-scaled clip)	97.99	**87.92**	**94.69**
v3	v2 + Elliptical Top-Hat / Bottom-Hat (11×11 kernel)	95.18	77.58	89.41
v4	v3 + Uncalibrated Macenko SVD Optical Density Transfer	57.78	42.28	52.69

Monocytes from Lymphocytes.

3. **Instability of Uncalibrated SVD (v4):** Applying Macenko optical density decomposition (Macenko et al., 2009) without slide-specific calibration drops accuracy to 57.78% on TestA and 42.28% on TestB. When erythrocytes and plasma dominate the patch, the SVD components follow background hemoglobin instead of the nuclear dye axes. Moment-based Reinhard transfer in CIE $L^*a^*b^*$ space does not have this problem.

5.8 Scenario 5: Explainability Guardrail and Clinical REST API Evaluation

Throughout training, the mean foreground focus ratio recorded by `XAIFocusMonitor` stayed between 0.64 and 0.79, above the 0.55 threshold, so the monitor never stopped a run. This is consistent with foreground cropping and background noise keeping the model off background shortcuts. At inference time, foreground-masked Guided Grad-CAM (α_{focus}) removed the halos over background plasma, so the LLM verifier could judge nuclear lobation and cytoplasmic granulation without false alarms from surrounding erythrocytes. Under concurrent load testing of the containerized Flask REST API (`/predict` and `/health` endpoints), invalid file uploads (`.pdf`, `.docx`) returned `HTTP 415 Unsupported Media Type`, corrupted byte streams returned `HTTP 400 Bad Request` without server failure, and valid microscopy requests had 0% packet drop with end-to-end latency between 200 and 350 ms (median GPU forward pass of 18.5 ms).

5.9 Discussion of Findings

High in-distribution accuracy says little about out-of-distribution stability. Vanilla DenseNet121 scored 98.57% on TestA and 73.24% on TestB, while our 7.83M-parameter model with inference adaptation reached 89.05% on TestB, within 0.66 pp of the 138M-parameter VGG16 (89.71%) with 5.7% of its parameters.

Separating covariate shift from label-space shift shows where the TestB gain comes from. Target subspace projection contributed 16.94 pp and Reinhard transfer with TTA contributed 15.15 pp. Color moment matching therefore corrects most of the camera-related color drift, which uncalibrated SVD normalization could not do (Scenario 4).

Finally, a monitor that checks attribution during training (`XAIFocusMonitor`) can stop a run that drifts toward background features. Post-hoc visualization can only reveal such a shortcut after the model has learned it.

6 Conclusions and Future Work

A smear classifier used in clinical laboratories has to cope with different microscopy hardware, background shortcuts, class imbalance, and its own opacity. We built a leukocyte classification pipeline for these conditions on a 7.83M-parameter DenseNet121 backbone. The system includes a five-stage Medical Enhanced Filter (MEF) to standardize exposure and contrast, a channel-spatial attention module (*WBCAttention*), learnable *MedSwish* activations, and an online training monitor (`XAIFocusMonitor`) that checks attribution against cell masks. At test time, Reinhard stain matching, test-time augmentation, and target subspace projection raised cross-device accuracy on TestB to 89.05%, from 56.96% for the unadapted five-class model and 73.90% after subspace projection alone. Of the 32.09-point gain, 15.15 points came from visual adaptation. On TestA the model reached 98.53% accuracy, with an F_1 of 1.0000 on Basophils, at a median of 18.5 ms per image on the GPU.

The work has three limitations. First, all experiments used pre-cropped cell patches from Giemsa-stained smears; full whole-slide image analysis requires an upstream cell detector, and other stains (such as Wright or May-Grünwald) need their own reference color pro-

files. Second, the Raabin-WBC TestB split provides ground-truth labels only for Neutrophils and Lymphocytes, limiting five-class out-of-distribution evaluation on that test set. Third, querying cloud-hosted multimodal models (GPT-4o and Gemini 2.5 Flash) introduces external network dependencies and patient data privacy constraints under clinical data regulations.

Future work includes connecting the classification backbone to an object detector for whole-slide smears, evaluating calibrated stain dictionaries across multicenter cohorts, and testing locally hosted, quantized vision-language models to provide on-premise verification without cloud API dependencies.

Acknowledgments and Code Availability

The author thanks Dr. Fatma Akalın and the Department of Information Systems Engineering at Sakarya University for their academic guidance. The Raabin-WBC dataset was generously provided by Tehran University (Kouzehkanan et al., 2022). Source code, trained weights, and reproducible evaluation scripts are publicly available at https://github.com/yildirimemirhan/wbc-analyzer.

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